

CHEM 515 Practice Session 6.1: Molecular Dynamics simulations: system setup, execution, and trajectory analysis

Name: _____

In this practice, you will practice (i) setting up molecular dynamics (MD) simulations for a protein-ligand complex using AmberTools, and (ii) the full process of MD simulation for a solute-solvent system (HBDI⁻ in water) from parameterization and system setup to analysis of the MD trajectory. You will also practice using the VMD software for visualization and analysis. Note that the procedure involved in **Exercise 1** is pretty much the same as the second example demonstrated in Nov. 19's class.

You will need the AmberTools MD software package to complete the two exercises below, which has been installed on the Ictotck server. If you haven't added the Amber script to your `$HOME/.bashrc`, please do the following:

```
echo "source /opt/amber22/amber.sh" >> $HOME/.bashrc
source $HOME/.bashrc
```

After you do these, you will get access to the programs under AmberTools (you don't need to do this again next time you log in). You can check if you have successfully done the setup by typing "`which sander`" on your command line and hit **Enter**: you should see "`/opt/amber22/bin/sander`" pop up.

You will need to complete the two exercises below following the instructions step-by-step and submit the files (including images and movies) that are requested. There are also a few simple questions for you to complete on this handout. **7.5 points** are allocated for the completion of this handout, **1 point** for attendance, and the remaining **1.5 points** are reserved for an additional quick assignment to be announced next Tuesday (which is related to AlphaFold3).

Exercise 1: Setting up the MD simulation for a protein-ligand complex using AmberTools

In this exercise, we will use Ambertools to prepare the MD simulation for the complex between sustiva (a small-molecule drug serving as an inhibitor) and the protein structure of the HIV-1 reverse transcriptase (RT), which we refer to as the **sustiva-RT complex** below. Upload the provided PDB file `1FK0_trunc.pdb` to the Ictotck server, then you can set up the MD simulation following the procedure below:

1. We start from processing the original PDB file using the `pdb4amber` script provided in AmberTools. This involves the following two commands:

```
pdb4amber -i 1FK0_trunc.pdb -o 1FK0_pro_dry.pdb -p --dry --add-missing-atoms
pdb4amber -i 1FK0_pro_dry.pdb -o 1FK0_pro_processed.pdb -y
```

In the first command, "`-p`" stands for "selecting protein only", "`--dry`" removes all the crystal waters, and "`--add-missing-atoms`" adds the missing heavy and H atoms. The second command then removes all the H atoms (through "`-y`") from the PDB file generated by the first command: by doing this, we leave the task of adding missing H's to the `tleap` program in a later step, which is recommended. The output PDB file of the second command, `1FK0_pro_processed.pdb`, will be used later by `tleap`.

2. A byproduct of the first `pdb4amber` command above is another PDB file "`1FK0_pro_dry_nonprot.pdb`", which contains the *non-protein* part of the original PDB file. By investigating this file, you will find that it simply contains the small-molecule ligand (labeled as "HETATM" with rename "EFZ"). This small molecule needs to be parameterized separately within the General Amber Force Field (GAFF) using the `antechamber` program. To make it ready for that, we first use the `reduce` command to add

the missing H's to the small molecule:

```
reduce 1FK0_pro_dry_nonprot.pdb > sustiva.h.pdb
```

Albeit not absolutely necessary, let's update the `resname` of the ligand from "EFZ" to "SUS". This can be easily achieved using a Linux command:

```
sed -i 's/EFZ/SUS/g' sustiva.h.pdb
```

3. Now we can use the `antechamber` program to parameterize the sustiva molecule:

```
antechamber -i sustiva.h.pdb -fi pdb -o sustiva.mol2 -fo mol2 -c bcc -s 2
```

Here "-i"/"-fi" and "-o"/"-fo" specify the input/output file name and format, respectively; "-c" specifies the scheme to generate the atomic charges, and here we choose the semi-empirical method AM1-BCC; "-s" specifies the verbose level of the output, which is not too important. The output file "sustiva.mol2" is what we need, which defines this small molecule. We can then use `parmchk2` to check the availability of the parameters in GAFF and generate the `.frcmod` molecule, which contains information about the force field parameters that the program is "less sure" about (typically assigned based on analogy to similar atomtypes):

```
parmchk2 -i sustiva.mol2 -f mol2 -o sustiva.frcmod
```

You should inspect the `sustiva.frcmod` file generated. If it contains no attention information ("ATTN"), then we are good to proceed.

4. You are now ready to set up the MD simulation for the sustiva-RT complex, that is, to generate the topology and input coordinates. While most of the time we execute `tleap` through an input file that contains a series of commands, as your very first practice, it is better to go through the procedure step by step. First, type "`tleap`" on your command line and then hit Enter. The commands you need to set up the simulation are provided below (**Note:** the lines starting with a "#" are **comments** which should NOT be executed):

```
# load the force field parameters for protein, small molecule, and water
source leaprc.protein.ff19SB
source leaprc.gaff
source leaprc.water.opc

# load the sustiva ligand
loadamberparams sustiva.frcmod
SUS = loadmol2 sustiva.mol2
check SUS

# load the protein (RT)
RT = loadpdb 1FK0_pro_processed.pdb
check RT

# form the sustiva-RT complex
complex = combine {RT SUS}
# solvate, then add counterions
solvateOct complex OPCBOX 12.0
addIons complex Na+ 0
addIons complex Cl- 0
# save the coordinates and topology files
saveamberparm complex sustiva_RT_solvated.prmtop sustiva_RT_solvated.inpcrd
savepdb complex sustiva_RT_solvated.pdb
# we are done!
quit
```

We are done with the setup now! The generated topology (`.prmtop`) and input coordinates (`.inpcrd`) files are what we need to proceed with Amber MD simulations.

We are now going to use VMD to visualize the solvated sustiva-RT complex we just set up. Open up VMD, select "File → New Molecule", and then load the PDB file "sustiva_RT_solvated.pdb" saved by

`tLeap` (in this case, VMD should be able to automatically detect the file type and change it to “PDB”). Let’s first turn off the axis (“Display → Axes → Off”) and change the view from “Perspective” to “Orthographic” (also under “Display”). Then we are going to create separate representations for the protein, ligand (`resname SUS`), and Cl^- ions (`resname "Cl-"`) under Graphics → Representations. The details are specified in **Table 1**:

Table 1: Representations needed in the VMD visualization of the solvated sustiva-RT complex

Selection	Drawing Method	Coloring Method
protein	NewCartoon	Secondary Structure
resname SUS	CPK	Name
resname "Cl-"	VDW	Color ID 7 (green)

For the VDW representation for Cl^- , increase the “Sphere Scale” to **1.5**, and set “Sphere Resolution” to **above 30**. Now you can save the image in the OpenGL Display window via File → Render: the `.tga` format is okay for submission but ideally you should further convert the `.tga` file to a `.pdf` or `.png` file.

Questions to answer:

1. Is the simulation we set up an explicit solvent or implicit solvent simulation? What will be the shape of the periodic box used in the simulation we set up? (**0.25 pt**)
2. How many missing atoms have been added to the system by `tLeap`? (**0.25 pt**) What type of ions have been added and how many? (**0.25 pt**) (**Hint:** you should be able to find answers to these from the `leap.log` file, if you didn’t pay much attention when executing the `tLeap` commands.)
3. The small-molecule drug sustiva contains fluorine atoms. Based on the AM1-BCC parameterization, are the F atoms the most negatively charged atoms in this molecule? (**0.25 pt**) (**Hint:** the atomic charges can be found in the rightmost column under the “ATOM” section of the “`sustiva.mol2`” file.)

Files to submit:

1. The input coordinates and topology files you generated using `tLeap` (**0.5 pt**)
2. Visualization of “`sustiva_RT_solvated.pdb`” generated using VMD based on the given specifications in **Table 1** (**1 pt**)

Exercise 2: MD simulation of HBDI^- in water

HBDI (*p*-hydroxybenzylidene-2,3-dimethylimidazolinone) is a synthetic analog of the chromophore in the green fluorescent protein (GFP), whose anionic form (HBDI^-) is shown in **Fig. 1**. In this exercise, we will

1. Generate the force field parameters for HBDI^- and set up the MD simulation for a single HBDI^- molecule solvated in water

2. Complete the energy minimization and heating steps with the input coordinates and topology files you generate and the provided input files for the **sander** program
3. Perform several trajectory analysis tasks based on a provided trajectory obtained from a production MD simulation

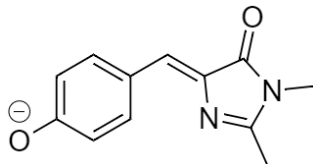


Figure 1: The molecular structure of HBDI⁻

Here we start from a DFT-optimized HBDI⁻ structure in the XYZ format (“HBDI-**optimized.xyz**”). You can open it up using IQmol, and then click “File → Save as”. Make sure that you change the file format to “.pdb”, and we can simply name it as “HBDI-**iqmol.pdb**”. Upload this file to the ITCOTCK server. We can now use the **pdb4amber** program to process this PDB file to make it fully compatible with Amber:

```
pdb4amber -i HBDI-iqmol.pdb -o HBDI-amb.pdb
```

You may have noticed that in the original PDB file the resname for HBDI⁻ is “UNL”, which stands for “unknown ligand”. Let’s change it to “CRO” (which stands for “chromophore”) using the same Linux command (“sed”) as in **Exercise 1**:

```
sed -i 's/UNL/CRO/g' HBDI-amb.pdb
```

You are ready to use **antechamber** to generate the force field parameters:

```
antechamber -i HBDI-amb.pdb -fi pdb -o HBDI-.mol2 -fo mol2 -c bcc -nc -1 -s 2
```

Note that an additional keyword “-nc -1” is needed here, which sets the charge of the small-molecule species to -1 (“nc” stands for “net charge”). Then, similar to the last example, let’s use **parmchk2** to check the availability of the force field parameters in GAFF:

```
parmchk2 -i HBDI-.mol2 -f mol2 -o HBDI-.frcmod
```

Examine the “HBDI-.frcmod” file obtained by opening it up using **vi**. You will notice that it contains more lines this time with a few items of pretty high *penalty score* (mostly for dihedrals), which renders a caveat that some of these force field parameters may *not* be very accurate. Nevertheless, for this exercise we can proceed without paying extra attention to these as long as the program is able to assign force field parameters from GAFF.

Now we are ready to use the **tleap** program to complete the MD setup. In this exercise, we will execute **tleap** with an input file, which is more convenient in practice since you can easily start from a template. Here a template for the **tleap** input file (“HBDI-**leap.in**”) has been provided under the “HBDI-**water_md**” directory. You need to **complete the following lines** in the file based on the instructions provided by the comments:

```
# Solvate in a 12.0 Angs TIP3P truncated octahedron box
[to be completed...]

# Add Na+ / Cl- to neutralize
[to be completed...]
```

In general, these commands should be *very similar* to what we used in **Exercise 1**, except that the name for the water box needs to be changed to “TIP3PBOX” here. After you fill in these lines, simply execute

```
tleap -f HBDI-leap.in
```

to complete the MD setup.

As in the previous exercise, you are supposed to use VMD to visualize the PDB file saved by `tLeap` (“HBDI-`water`.pdb”). The visualization should contain the representations specified in **Table 2**:

Table 2: Representations needed in the VMD visualization of the solvated HBDI⁻

Selection	Drawing Method	Coloring Method
resname CRO	CPK	Name
resname “Na\+”	VDW	Color ID 27 (magenta)
water	line	Name

Again, turn off the “Axes” and change the view to “Orthographic”. Save the generated visualization as a figure, which needs to be submitted with your work.

Now we are ready to proceed with MD simulations with the prepared input coordinates and topology files. As we introduced in class, the complete procedure for an MD simulation typically involves (i) energy minimization, (ii) heating and equilibration, and (iii) production run. With AmberTools, all these steps will be completed using the `sander` program. In this exercise, **you only need to complete the minimization and heating steps** with the given input files for `sander` (“`min.in`” and “`heat.in`”); the output (“`prod.md.out`”) and trajectory (“`prod.md.crd`”) files obtained from the production MD run has been provided, based on which you will need to complete a few trajectory analysis tasks. First, we minimize the energy of the system with command:

```
sander -O -i min.in -o min.out -p HBDI-water.prmtop -c HBDI-water.inpcrd -r min.rst &
```

Here “`-i`” and “`-o`” specify the names of the input and output files, respectively, “`-O`” means overwriting the output file (“`min.out`”) if it already exists, “`-p`” and “`-c`” specify the names of the topology and input coordinates files, respectively, and “`-r`” specifies the name of the *restart* file, which in this case will just be the final structure obtained from the energy minimization.

From the input file “`min.in`”, you should be able to tell that it is doing 3,000 minimization steps in total, including 1,000 steepest descent steps followed by 2,000 steps of conjugate gradient optimization. For the full explanation of the keywords involved in “`min.in`”, please refer to the [alanine dipeptide tutorial](#) on the Amber website (under “**Minimization input**”).

After the minimization is done (which should only take ~1 min), open up the “`min.out`” file and report the energy for the initial and final steps (**0.2 pt**):

- Initial energy (NSTEP = 1):
- Final energy (NSTEP = 3000) :

Note that the energies in Amber outputs are in kcal/mol. By skimming through the energy values printed out every 100 steps, is the system energy changing in the right direction during the **energy minimization** process? (**0.3 pt**)

We then perform a relatively short MD simulation to (i) elevate the temperature of the system from 0 K to 300 K and (ii) let the system equilibrate at 300 K. The last 3 lines of the input file “`heat.in`” specifies that in the first 10,000 MD steps (20 ps with a time step of 0.002 ps), the temperature will be raised from 0 to 300 K, and then the system will equilibrate at 300 K for another 15,000 time steps (30 ps). Regarding the details of the input keywords, please refer to the [online tutorial](#) again (under “**Heating input**”). The command you need to execute for the heating step is:

```
sander -O -i heat.in -o heat.out -p HBDI-water.prmtop -c min.rst -r heat.rst -x heat.nc &
```

which could take 8–10 minutes to finish on Ictotck. Here we set “`-c min.rst`” so that the input coordinates for the heating step come from the restart file generated by the energy minimization step, and the new keyword “`-x`” specifies the output trajectory file: with 25,000 time steps in total and “`ntwx = 250`”

(saving coordinates every 250 steps) in `heat.in`, the resulting trajectory file `heat.nc` should contain 100 MD snapshots.

The trajectory file `heat.nc` is a binary file which cannot be read directly using `vi` or other text editors, and it may not be supported by VMD either on some operating systems. To visualize and analyze the heating trajectory using VMD, let's first convert `heat.nc` into a text trajectory file. This can be done using the `cpptraj` program in AmberTools:

```
cpptraj -p HBDI_water.prmtop -y heat.nc -x heat.crd
```

You can now copy `heat.crd` back to your computer. To visualize it using VMD, under “File → New Molecule” you need to first load the topology file `HBDI_water.prmtop` (make sure that the file type is shown as “AMBER7 Parm”). Now the top line in the window should change to “Load files for 0:HBDI_water.prmtop”: click “Browse”, select “`heat.crd`” from the location you save this file, and make sure to **change the file type** to “AMBER Coordinates with Periodic Box” (**important**). The file loading window should now look like in **Fig. 2**:

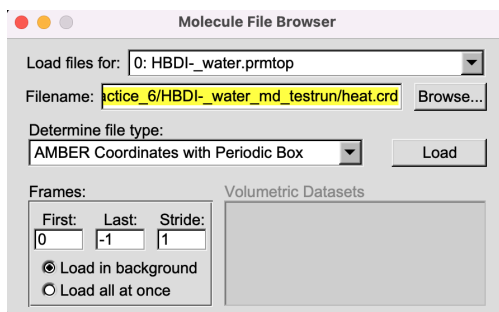


Figure 2: Loading Amber trajectory to VMD

Now you can click “Load” to load the trajectory in VMD. You may notice that there is an extra bond connecting the two H’s of each water, which is because VMD regards TIP3P as a rigid water model. We don’t need to deal with that right now. Instead, let’s create the representations for the HBDI[−] chromophore (`resname CRO`) and the Na⁺ cation (`resname "Na\+"`) only with the same specifications in **Table 2**, and then set “Mouse → Label → Bonds” and click on the Na⁺ cation and the oxygen atom on the **phenolate ring** (the 6-member ring) of HBDI[−]. You should be able to see two atom labels and a “bond” showing up on the OpenGL Display. Play the trajectory with a slower speed and check out how this distance changes over the entire trajectory.

To get a quantitative measure, go to “Graphics → Labels” and select “Bonds” from the dropdown menu; then click on the bond between “CRO1:O” and “Na+2:Na” and switch to “Graph” tab, and click “Graph”. This will generate a plot of this distance over time. **Save this plot**, which needs to be submitted with your work. **Note:** you can first save it as a “.ps” file, and that can be converted to a PDF using Adobe Acrobat Distiller or other software apps. If you are having troubles converting it to a PDF file, you can also do “File → Export to ASCII vectors”, which will produce a “.dat” file that contains two columns of data. You can then import the data into Excel and generate a graph.

For the production MD, the input, output, topology, and text coordinates files have been provided. While **you are NOT supposed to do the production run yourself** (since it is going to take a few hours), the command for that is as follows:

```
sander -O -i prod_md.in -o prod_md.out -p HBDI_water.prmtop -c heat.rst -r prod_md.rst -x prod_md.nc &
```

From the “`prod_md.in`” file, one can tell that the production run is an **NPT simulation at 300 K**. The full explanation for the keywords involved can be found in the same [tutorial](#) online (under “**Production input**”). Based on the total number of time steps (given by “`nstlim`”) and the size of the time step (“`dt`”), **what is the total time duration of this simulation?** Show your calculation below and report your result in ns (1 ns = 10³ ps = 10⁶ fs): **(0.25 pt)**

Now let's move to the trajectory analysis part. **Start a new VMD session** and load the trajectory "prod_md.crd" following the same procedure as introduced above (**Note:** the topology file to load here should be the provided "prod_md.prmtop" file rather than your own "HBDI-water.prmtop"). Using the same procedure as above, generate the plot of the O(phenolate)···Na⁺ distance over the entire production trajectory and **save the figure**. **Describe what you observed** from this plot and its difference from that generated from the heating trajectory: **(0.5 pt)**

Let's now investigate the root-mean-square displacement (RMSD) of the HBDI⁻ solute (excluding H's) over the production MD run. Go to "Extensions → Analysis → RMSD Trajectory Tool", change the selection in the first text box from "protein" (which we don't have here) to "resname CRO" and leave the "noh" (no hydrogen) option selected. On the right of the same window, make sure that "Frame ref:" is 0 and select "Plot" as well.

To generate the so-called "best-fit" RMSDs, you should first click "ALIGN" to let the program overlay the selected part in each frame on top of that in the reference frame as much as possible (note: this will change how the trajectory looks as well), and then click "RMSD". A plot of the RMSD against the MD frame number will be generated. **Save this plot**, which needs to be submitted with your work.

From the RMSD plot, you should be able to notice that there is a sudden "burst" roughly starting from frame 200. This may indicate some unusual event in the MD simulation. To further investigate that, first let's delete all the labels we have created so far under "Graphics → Labels → Atoms" and "Graphics → Labels → Bonds". Then we create a new label for one of the two H atoms at the *meta*-position of the phenolate group (using "Mouse → Label → Atoms"). Play the trajectory from frame 190 with a slower speed and watch how the position of the labeled H atom changes.

Based on your observation and the definition of RMSD, explain what results in this sudden increase of RMSD after frame 200 **(0.25 pt)**. Then, based on the molecule structure of HBDI⁻, discuss how likely or unlikely it is for this "event" to occur and its implications for the force field parameters we generated for this molecule. **(0.5 pt)**

As the final task, let's **create a movie** for the production MD trajectory. First, delete the axes (if you still have it) and the atomic label you just created, and make sure that it is under the **Orthographic** view. You should also make sure you have **aligned** your trajectory with respect to the first frame (you have done this if you just made the RMSD plot under the same VMD session). Then you should roughly center the HBDI⁻ molecule on the OpenGL Display, and then add the water molecules back. Since the TIP3P water has the weird H–H connection, we can use the representations in **Table 3** for water:

Table 3: Representations for TIP3P water in VMD

Selection	Drawing Method	Coloring Method
name O H1	Line	Name
name O H2	Line	Name

Optionally, you can change the background color to "15 iceblue". The OpenGL Display should now look

like that shown in **Fig. 3**. Now go to “**Extensions** → **Visualization** → **Movie Maker**” to create a movie for this trajectory. If you are unfamiliar with the movie-making procedure, please refer to the recording of **Lecture 18**. Two things to note:

- Remember to set the working directory in the movie-making window (otherwise you may have trouble finding the movie file generated)
- Under “**Movie Settings**”, you should select “**Trajectory**”

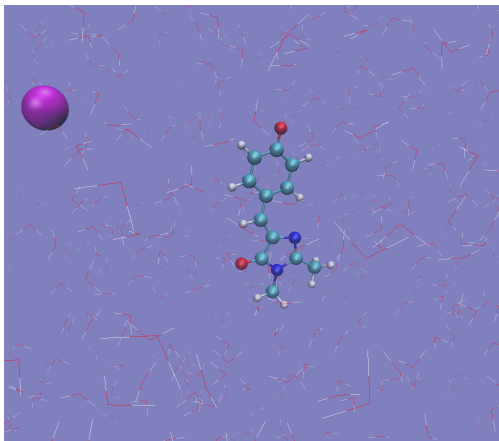


Figure 3: VMD view for the movie generation

Files to submit:

1. The completed `tleap` input file “`HBDI-leap.in`” (**0.2 pt**) and the prepared input coordinates and topology files (`HBDI-water.inpcrd` and `HBDI-water.prmtop`) (**0.2 pt**)
2. VMD visualization of the MD setup (the “`HBDI-water.pdb`” file generated by `tleap`) with the given specifications (**0.5 pt**)
3. VMD-generated plots of the $\text{O} \cdots \text{Na}^+$ distance over (i) the heating trajectory and (ii) the production MD run. (**1 pt**)
4. VMD-generated plot for the RMSD of the heavy atoms in HBDI^- during the course of production MD run (**0.5 pt**)
5. VMD-generated movie for the production MD run (**0.6 pt**)